EX. NO: 02

DATE: 29/07/2026

SOLVE A GRAPH USING DEPTH FIRST SEARCH

AIM:

To implement DFS traversal on a graph to explore all nodes from a source.

PYTHON CODE:

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
    visited.add(start)
    print(start, end="")
    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)
graph = {
'A': ['B','C'],
'B': ['D','E'],
'C': ['F'],
'D': [],
'E': ['F'],
'F': []
}
print("DFS traversal from node A:")
dfs(graph, 'A')
```

OUTPUT:

```
DFS traversal from node A:
ABDEFC
```

RESULT:

Thus the program of implementing DFS traversal on a graph to explore all reachable nodes from a source was executed successfully.